Does Asking Again Help? Majority Voting, Run-to-Run Consistency and Agreement of a Small Open Language Model on Oncology Exam Questions

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Studies that compare large language models (LLMs) with oncology tumor boards often query a model once per case, yet the same model can answer differently when asked again. Using oncology exam questions as a controlled testbed, we asked whether a majority vote over repeated sampled answers is more accurate than a single greedy answer for a small open-weight model, and how far agreement between runs signals correctness. We answered 633 oncology questions from the MedQA licensing-exam dataset with Qwen2.5-7B-Instruct, once with greedy decoding and ten times with sampling at temperature 0.7, under a protocol frozen before any model output on test items existed. The majority of the ten sampled answers was correct for 70.6% of questions and the greedy answer for 67.1%, a difference of 3.5 percentage points (95% CI 0.9 to 6.0; exact McNemar p = 0.012), at 10.0 times the output tokens; in a post hoc split, 2.6 points (1.1 to 4.0) came from voting itself rather than from sampling. Greedy decoding repeated its answer for 98.7% of questions, two sampled runs agreed for 78.6%, and all ten for only 49.4%. Agreement between runs tracked correctness (AUROC 0.737), but 13.1% of unanimous answers were wrong, and the runs shared a preference against one answer position. Voting gives a modest, uncertain gain for a small open model; consistency depends on the model and its decoding settings, which should be reported with it; and agreement is only a weak signal of correctness. Exam questions are a proxy for, not a test of, tumor-board decisions.

Additional Key Words and Phrases: large language models, oncology, exam questions, self-consistency, majority voting, reproducibility, MedQA

1 Introduction

AI systems, from rule-based decision support to large language models (LLMs), are increasingly evaluated against the decisions of multidisciplinary tumor boards [1, 10, 11, 19, 26], and several recent studies describe a single model recommendation per case [4, 15, 27]. Yet an LLM asked the same question again can answer differently: where tumor-board studies repeated their queries, agreement ranged from about a third of cases for ChatGPT [1, 11] to near-perfect for some models and poor for reasoning models [10], and on benchmarks consistency also varies with the temperature [12, 14]. Repetition can also be used: several sampled answers can be combined by a majority vote (self-consistency [21]), and agreement between runs can signal how far an answer can be trusted [18]. In oncology, voting has been tested with strong closed models and brought little benefit [7, 20, 23]; whether it helps a small open model, which varies more, is not known.

We use oncology questions from the MedQA licensing-exam dataset [8] as a controlled testbed with a single keyed answer, and compare greedy decoding with a majority vote over 10 sampled runs of Qwen2.5-7B-Instruct on 633 questions, under a protocol frozen before any model output on test items existed. We ask: (RQ1) Is the majority of 10 sampled runs more accurate than the greedy answer? (RQ2) How consistent are the answers across runs? (RQ3) Does agreement between runs flag answers that are more likely to be correct? Our contributions are:

- a pre-specified paired comparison in which the vote was 3.5 points more accurate than greedy decoding (95% CI 0.9 to 6.0) at 10.0 times the output tokens, 2.6 points of it, post hoc, attributable to voting itself;
- like-for-like consistency figures: greedy repeated its answer for 98.7% of items, two sampled runs agreed for 78.6%, all ten for 49.4%;

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- evidence that agreement tracks correctness (AUROC 0.737), with its limits: it partly reflects item difficulty, 13.1% of unanimous answers were wrong, and the runs share an answer-position preference;
- a frozen protocol, code, item identifiers and parsed answers from which every statistic can be recomputed.

2 Related Work

AI evaluated against tumor boards. Early rule-based systems such as IBM Watson for Oncology were compared with board recommendations [26]; more recent studies compare LLMs with board decisions in breast cancer [4, 11], lung cancer [15, 27], gynecologic oncology [1] and sarcoma [10], simulate the board with an LLM [20], or design multi-agent systems for multidisciplinary decision-making [25]. In earlier work we mapped this literature across rule-based, LLM and multi-agent approaches [19].

Run-to-run consistency. Responses of ChatGPT to the same breast-cancer cases asked three times were entirely consistent in only 32% of instances (abstract; the rule is not stated) [11]; with ChatGPT-5 asked three times per gynecologic case, the core recommendation was identical for 37.2% of 242 cases [1]. Both used default or unreported sampling settings and free-text answers. Li et al. [10] queried four models 21 times on five sarcoma cases, three at temperature 0: two answered almost identically, but a reasoning model did not, even at temperature 0. On MedQA and MMLU-Redux, small open models repeated ten times at three temperatures gave a stable answer (at least 9 of 10 runs) on about half to four-fifths of the questions, answering with a letter only [14]; in clinical text extraction with open models, higher temperature generally lowered agreement and majority voting raised it [12]. In oncology, exam answers varied across runs for every model tested [2], agreement of three replicates with stated confidence picked out more reliable answers [18], and flagship models agreed closely across temperatures on biomedical text [23] (abstracts). Our review recommended that such studies report how many times each case was run and how consistent the runs were [19].

Majority voting. Self-consistency replaces the single greedy chain of thought by several sampled ones and keeps the most consistent answer [21] (abstract). In oncology, voting did not improve GPT-4 on radiation-oncology physics questions [7], did not significantly change GPT-5's tumor-board recommendations [20], and gave marginal gains in biomedical text processing [23]. With small open models on medical question answering, self-consistency traded tokens for accuracy and enlarged the benefit of fine-tuning [3], and voting ensembles gave small gains [24] (abstracts). In the records we screened, no study isolated the voting effect for a small open model on oncology questions with reported decoding, or examined agreement as a correctness signal with reasoning outputs on such questions.

3 Methodology

3.1 Study design

We ran a paired, within-item comparison: every test item was answered by the same model under two decoding strategies, and each item contributes one correct-or-wrong outcome per strategy. The study answers three questions, fixed before any model output on test items existed: **RQ1** (confirmatory): does the majority answer of 10 sampled runs have higher accuracy than the greedy answer? **RQ2** (descriptive): how consistent are the answers across the 10 sampled runs? **RQ3**: does agreement between runs flag answers that are more likely to be correct? Figure 1 summarises the pipeline.

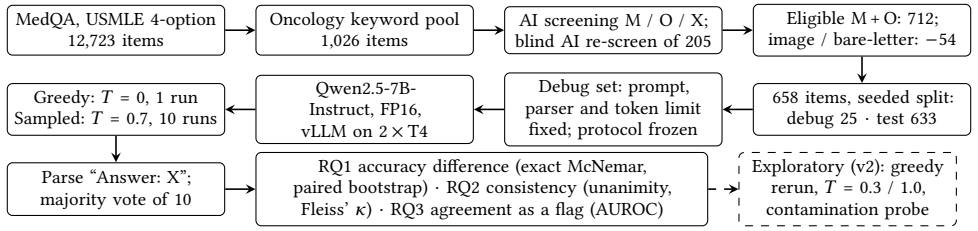


Fig. 1. Study pipeline. Item selection (top row, left to right), protocol freeze and generation (middle row, right to left), evaluation (bottom row). No model output on test items existed before the protocol was frozen.

3.2 Items

Items come from the English, four-option USMLE part of MedQA [8] on Hugging Face [6] (12,723 items, pinned commit). A keyword rule (term list in the repository's `count_oncology_items.py`) kept 1,026 items with an oncology term in the vignette or the correct answer. Since the pool was drawn from all of MedQA, the test set mixes items from its train (492), dev (70) and test (71) splits, which matters for contamination (Section 7). An AI screener (Claude Code) coded each item against written rules: *M*, managing a known or suspected neoplasm or a direct complication; *O*, a neoplasm tested otherwise (diagnosis, pathology, mechanism, risk, screening, adverse effects of cancer therapy); *X*, incidental. With no clinical checker available, a second, blind instance of the same model re-coded a random 20% sample (205 items) from the full text; they agreed on 190 (Cohen's $\kappa = 0.873$), and the first screener adjudicated the 15 disagreements, changing 8 codes (3.9%, our estimate of the miscoding rate among the 821 unsampled items). Of 712 eligible items (129 M, 583 O), 54 referring to an image or with bare-letter options were removed; a seeded split left 25 items for developing the prompt and parser and 633 test items (118 M, 515 O). An *objective subset* of 271 test items, whose correct answer itself contains an oncology term, depends less on screening judgement.

3.3 Model, prompt and decoding

We used Qwen2.5-7B-Instruct [16, 17] at a pinned revision, in 16-bit floating point, with its own chat template. The system message was *"You are a careful medical expert. Answer the multiple-choice question."*; the user message gave the question, the options A–D in the dataset's order, and the instruction to think briefly, step by step, and end with a line *"Answer: X"*. *Greedy decoding* used temperature 0 and one output per item; *sampling* drew 10 outputs per item at temperature 0.7, top-p 0.8 and top-k 20, the model's own defaults. Both used a repetition penalty of 1.05 and at most 512 output tokens; a truncated output is scored as wrong, which costs the single greedy answer more than the vote, so we also report a lenient analysis. Seeds were a fixed base plus the item's index; the first 50 test items were run greedily twice.

3.4 Answer extraction and majority vote

The answer is the last "Answer: X" in the output (Markdown bold allowed), or else the last "answer is X"; only a single capital A–D not followed by another letter counts. An output without one is *unparseable* and is scored as wrong. The *majority vote* is the most frequent letter among the parsed runs; unparseable runs do not vote, a tie goes to the tied letter that appeared first, and an item with no parsed run has no vote. The *modal share* is the number of runs giving the voted letter, divided by 10.

3.5 Outcomes and statistical analysis

RQ1. The primary outcome is the accuracy of the majority vote minus that of greedy decoding on the same items, tested with the exact two-sided McNemar test [13] on the discordant items (b: vote right, greedy wrong; c: the reverse), with a 95% paired percentile bootstrap CI (10,000 resamples). The decision rule was fixed in advance: *supported* if $p < 0.05$ *and* the lower bound is above 0; *practically meaningful* if the difference is at least 3 points. A power calculation for the then-planned 684 items gave a minimum detectable difference of 3.7–4.3 points at a nominal 80% power (77–78% in simulation with the exact test) for 12–16% discordant items. We also report the mean accuracy of a single sampled run and of a majority over the first k runs (k = 1, 3, 5, 7, 9, 10).

RQ2 and RQ3. Consistency is described (without a test) by the share of unanimous items, the mean modal share, Fleiss' κ [5] over the categories A–D and unparseable, and the pairwise flip rate (the share of an item's 45 run pairs that disagree), with bootstrap CIs (2,000 resamples for κ and the AUROC, 10,000 otherwise). For RQ3, the AUROC of the modal share for a correct vote is reported; H3 is *supported* if its lower bound exceeds 0.5, a deliberately modest bar. We also compare accuracy at ≥9/10 and <9/10 agreement and plot accuracy against coverage.

Sensitivity. RQ1 is repeated with unparseable greedy items dropped, on the objective subset, and in the M and O subgroups (M is descriptive only). Discordant items and items wrong in all 11 generations are analysed. Only RQ1 is confirmatory; secondary analyses are not corrected for multiple comparisons.

3.6 Protocol freeze, amendment and post hoc analyses

The protocol and its decision rules were frozen and committed to the project's repository (commit `7890add`) before any model output on test items existed, and acceptance criteria for the greedy baseline were written before the test run. The AI assistant that designed the protocol had read the test items and keys during screening, but no model output. After validating the baseline, and before looking at the voting comparison, we added an exploratory amendment as a new protocol version: a greedy rerun of all test items, 10 sampled runs at T = 0.3 and 1.0, and a contamination probe in which the first wrong option of 100 random test items was hidden and the model asked to reproduce its exact text (Wilson interval [22]). Analyses added after the results were seen, most of them at the request of a reviewer, are labelled *post hoc*.

4 Experimental Setup

Generation ran on Kaggle's free tier with two NVIDIA T4 GPUs (16 GB each), using vLLM 0.32.0 [9] with PyTorch 2.13, tensor parallelism over both GPUs and a context length of 2,048 tokens. Both arms were generated in batches of 64 items, so one engine call held 64 sequences for greedy decoding and 640 for sampling; GPU time therefore depends on this configuration, and we use output tokens as the measure of cost. Generation took 0.57 h for the main run and 0.95 h for the exploratory run (0.76 and 1.11 h of GPU session). Each notebook installed pinned versions, cloned the repository at a fixed commit, checked dataset checksums and ran the unit tests before generating. Evaluation ran on a CPU (Python 3.12, pandas 3.0, NumPy 1.26, SciPy 1.17), after an audit of the saved outputs (split checksums, no question shared between debug and test sets, prompts independent of the answer key, identical items, engine, model revision, seeds and token limit in every arm, chance accuracy with shuffled keys; 45 of 45 checks passed). Re-running the evaluation on another machine reproduced the Kaggle result files byte for byte. Greedy decoding on GPUs is not exactly deterministic (48 of 50 letters were the same in the 50-item rerun) and sampled outputs are not bit-reproducible. The raw outputs restate licensed question text and are

Table 1. RQ1: accuracy (%) of the majority of 10 sampled runs (T = 0.7) and of greedy decoding on the same items. Δ = majority minus greedy in percentage points, 95% paired bootstrap CI; b/c = items right only with the vote / only with greedy; p = exact McNemar test. Only the first row is confirmatory.

Analysis	n	Greedy % (95% CI)	Majority-10 % (95% CI)	Δ points (95% CI)	b/c	p
All items (confirmatory)	633	67.1 (63.5, 70.8)	70.6 (67.0, 74.2)	3.5 (0.9, 6.0)	46/24	0.012
Lenient parsing	625	68.0 (64.3, 71.7)	70.9 (67.2, 74.4)	2.9 (0.3, 5.4)	42/24	0.036
Objective subset	271	64.9 (59.4, 70.5)	71.2 (65.7, 76.8)	6.3 (2.2, 10.7)	26/9	0.006
Management (M)	118	66.9 (58.5, 75.4)	72.0 (63.6, 79.7)	5.1 (−1.7, 11.9)	11/5	0.21
Other oncology (O)	515	67.2 (63.1, 71.3)	70.3 (66.4, 74.2)	3.1 (0.4, 5.8)	35/19	0.040

Table 2. Post hoc analyses of the RQ1 difference: its split into a sampling part and a voting part, the vote against each single run and against the second greedy run, and the difference by the MedQA split each item came from.

Comparison	n	Δ points (95% CI)	b/c	p
Single sampled run (mean of 10) − greedy	633	0.9 (−1.2, 3.1)	–	–
Majority of 10 − single sampled run (mean)	633	2.6 (1.1, 4.0)	–	–
Majority of 10 − each single run (range)	633	1.9 to 3.6	39–49/21–35	0.007–0.17
Majority of 10 − second greedy run	633	4.1 (1.6, 6.6)	–	0.003
Majority − greedy, items from the MedQA train split	492	3.5 (0.4, 6.5)	37/20	0.033
Majority − greedy, items from the MedQA dev split	70	2.9 (−4.3, 11.4)	5/3	0.73
Majority − greedy, items from the MedQA test split	71	4.2 (−1.4, 11.3)	4/1	0.38

not released; every statistic can be recomputed exactly from the released parsed answers, and every number in the text is inserted by a script.

5 Results

Output quality. Every arm answered all 633 test items. 8 of 633 greedy outputs (1.3%) and 40 of 6,330 sampled outputs (0.6%) had no parseable answer; 7 and 36 of these had reached the 512-token limit, and the remaining 1 and 4, read during baseline validation, declined to choose an option. 25 votes were decided by the tie rule.

5.1 RQ1: majority vote versus greedy decoding

The majority vote answered 70.6% of the test items correctly (95% CI 67.0–74.2) and greedy decoding 67.1% (63.5–70.8), a difference of 3.5 percentage points (95% CI 0.9 to 6.0; Table 1); the vote alone was right on 46 items and greedy alone on 24 (exact McNemar p = 0.012). Under the pre-specified rule, H1 is *supported* and the point estimate passes the 3-point threshold, but the interval includes smaller gains, and with only 11.1% discordant items (12–16% were assumed for power) a significant estimate may overstate the gain. The direction was the same in every sensitivity analysis: 2.9 points with unparseable greedy items dropped, 6.3 on the objective subset, 3.1 in other-oncology items and 5.1 (−1.7 to 11.9) in the 118 management items, the closest proxy for treatment decisions.

The comparison changes decoding and aggregation at once. In a post hoc split (Table 2), a single sampled run was 0.9 points more accurate than greedy decoding (−1.2 to 3.1) and the vote 2.6 points more accurate than a single run (1.1 to 4.0), so most of the gain is due to voting; it was similar for items from MedQA's train, dev and test splits, and the vote also beat a second greedy run. Averaged over 200 random run orders (post hoc), majority-of-k accuracy rose smoothly from 68.0% (k = 1) to



Fig. 2. Majority of k sampled runs. Left: accuracy for the first k runs (protocol) and averaged over 200 random run orders with the 2.5–97.5 percentile band (post hoc); dashed line: greedy decoding. Right: output tokens per item.

69.8% (3), 70.5% (5) and 70.9% (10), whereas the protocol's first-k curve depends on the run order (Figure 2). The 10 runs used 10.0 times the output tokens of greedy decoding (2,491 vs. 248 per item).

5.2 RQ2 and RQ3: consistency and agreement

Measured like for like, greedy decoding repeated its letter for 98.7% of items, whereas two sampled runs at $T = 0.7$ agreed for 78.6%; all ten agreed for only 49.4% (Table 3), with Fleiss' κ 0.715. The distribution of modal shares was broadly similar for management and other items (Figure 3b).

The modal share separated correct from wrong votes with an AUROC of 0.737 (95% CI 0.695–0.780), so H3 is *supported*. When at least 9 of 10 runs agreed (406 items) the vote was correct for 83.5%, otherwise (227 items) for 47.6%; answering only unanimous items would cover 49.4% of the test set at 86.9% accuracy (Figure 3a). The signal needs many runs and partly reflects item difficulty (post hoc, Table 3): with only the first 3 runs the AUROC was 0.644, whether greedy agreed with the vote gave 0.586, and the 10-run modal share predicted the *greedy* answer's correctness about as well (0.749). 41 of the 313 unanimous items (13.1%) were unanimously wrong.

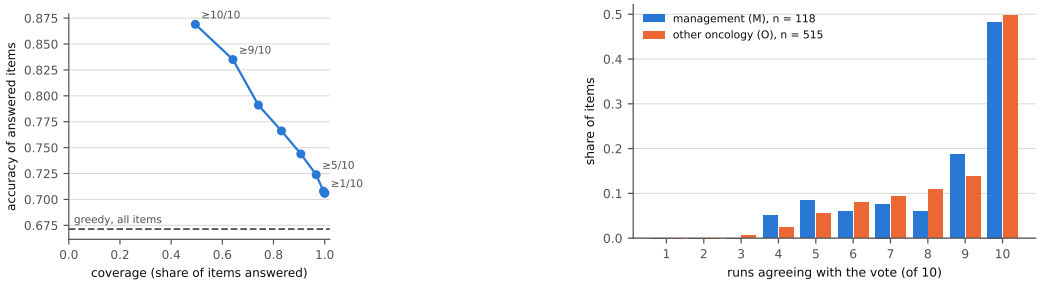
Shared answer-position preference (post hoc). Keys were spread evenly over A–D, yet wrong answers rarely chose A: 34 of 200 wrong greedy answers (45.3 expected; $p = 0.10$) and 315 of 1,982 wrong sampled answers (463.3 expected). Accuracy was lowest when the key was A (greedy 56.3%, vote 60.9%, against 68.1–73.2% and 71.9–75.0% otherwise). A preference shared by all runs correlates their errors, which limits what a vote can correct; it may also reflect item content.

5.3 Exploratory analyses (protocol amendment)

A second greedy run of all items scored 66.5%, −0.6 points from the first (−1.6 to 0.2). At $T = 0.3$ the vote reached 69.5% and gained 2.4 points over greedy; its bootstrap interval (0.2 to 4.6) excludes zero but the McNemar test does not ($p = 0.063$), so it would not meet the frozen rule, which requires both. At $T = 1.0$ the vote reached 70.9%, a gain of 3.8 points (0.9 to 6.6; $p = 0.013$). Unanimity was 54.0%, 49.4% and 47.6% at $T = 0.3$, 0.7 and 1.0; in a post hoc paired comparison it was lower at 1.0 than at 0.3 by −6.5 points (−9.6 to −3.5), while 0.7 and 1.0 did not differ clearly (−5.2 to 1.4). The AUROC stayed between 0.721 and 0.748. The contamination probe reproduced the hidden option exactly for 4 of 100 items (Wilson interval 1.6–9.8%), 3 of them among the 54 options of one or two words, which can be guessed.

Table 3. RQ2 and RQ3 for the 10 sampled runs at $T = 0.7$ (633 items); percentile bootstrap CIs over items.

Measure	Estimate (95% CI)
Unanimity, 10/10 runs (%)	49.4 (45.5, 53.2)
Mean modal share	0.858 (0.843, 0.872)
Fleiss' κ across runs	0.715 (0.689, 0.738)
Pairwise flip rate (%)	21.4 (19.5, 23.3)
AUROC, modal share vs. vote correct	0.737 (0.695, 0.780)
Vote accuracy, ≥9/10 agree (n = 406) (%)	83.5
Vote accuracy, <9/10 agree (n = 227) (%)	47.6
Coverage / accuracy at 10/10 agreement (%)	49.4 / 86.9
Post hoc comparisons	
Pairwise agreement between two sampled runs (%)	78.6
AUROC, modal share of first 3 runs vs. vote correct	0.644 (0.604, 0.685)
AUROC, greedy agrees with vote vs. vote correct	0.586 (0.551, 0.621)
AUROC, modal share vs. greedy correct	0.749 (0.709, 0.788)



(a) (b)

Fig. 3. (a) Accuracy against coverage when only items with at least x of 10 agreeing runs are answered (dashed: greedy on all items). (b) Runs agreeing with the vote, for management (M) and other oncology (O) items.

5.4 Error analysis

By construction, on the 46 items where only the vote was right greedy's letter was not the majority letter; the informative part is how rare it was: it appeared in 2.7 of the 10 sampled runs on average (for the 42 items with a parsed greedy answer), while the correct letter appeared in 6.8. On the 24 items where only greedy was right, the correct letter appeared in 3.0 of 10 runs. 5 and 6 of these items, respectively, were decided by the tie rule. 78 items were answered wrongly in all 11 generations: in 41 every generation chose the same wrong option (these are exactly the unanimously wrong items), 35 had two or more different wrong options, and 2 mixed a wrong option with unparseable output.

As an anecdotal check, the AI assistant read two random items per category: greedy sometimes named the right finding but chose another option; runs sometimes split between two plausible options; in one item several runs declined to choose; and in always-wrong items every run reasoned towards a non-keyed diagnosis or mechanism. No clinician checked these readings.

6 Discussion

A modest, uncertain gain, mostly from voting. The majority of 10 sampled answers was more accurate than a single greedy answer; the gain is statistically supported and its point estimate passes

the threshold set in advance, but the data are compatible with any gain from 0.9 to 6.0 points, it costs 10.0 times the output tokens, and most of it comes from aggregating runs rather than from sampling itself. Self-consistency was proposed on the intuition that several reasoning paths converge on the correct answer more often than a single one [21]; our data fit this view, since greedy's wrong answers were rare among the sampled runs, but do not prove a mechanism. A second greedy run, parse failures, ties and the MedQA split of origin do not change the direction. Strong closed models gained little from voting in oncology [7, 20, 23]; a model that is already consistent has less to gain, which may explain the difference, but with one model tested this remains a hypothesis.

Consistency depends on the model and its decoding. Greedy decoding repeated its answer for 98.7% of items, two sampled runs agreed for 78.6%, and unanimity fell between $T = 0.3$ and 1.0. That consistency varies with temperature was shown before for small open models [14] and clinical text extraction [12], and that it varies between models in a tumor-board study [10]; we add the same picture for reasoning outputs on oncology questions. Our figures cannot be ranked against the ChatGPT tumor-board studies [1, 11] (real cases, free text, three runs, unreported settings), and our 9-of-10 stability (62.2–67.0%) is only loosely comparable with letter-only answers [14]. The number of runs and their consistency, which our review recommended reporting [19], can only be interpreted together with the decoding settings.

Agreement is a weak signal. Agreement tracked correctness (AUROC 0.737), but it needs many runs, partly reflects item difficulty, and 13.1% of unanimous answers were wrong, all of them also wrong under greedy decoding. Such systematic errors, presumably gaps in knowledge or reasoning, and the runs' shared preference against option A are invisible to both voting and agreement.

Implications. These results concern exam questions with a single keyed answer, not board decisions, and say nothing about patient care or clinical use of model output. For studies that compare LLMs with tumor boards they support one reporting practice: state how many times each case was run, with which decoding settings, and how consistent the runs were.

7 Limitations

Exam questions are a proxy. MedQA items have one keyed answer and were written for a licensing exam; a tumor board weighs imaging, pathology, staging, comorbidity, guidelines and the patient's wishes, often accepts more than one option, and decides by discussion. Only 118 of our 633 items concern management, and their result is imprecise. Nothing here measures clinical usefulness or safety.

Contamination. Most test items (492 of 633) come from MedQA's training split, which is widely redistributed; whether the model saw them is unknown. The probe is weak (one hidden option, exact match, no control set, many short options). Memorised items would raise accuracy and could make runs agree more, which would shrink, not protect, the voting gain; similar gains in items from MedQA's train, dev and test splits argue against a large effect, but those strata are small.

Screening by AI. Oncology relevance was coded by an AI model and checked on a sample by a second, blind instance of the same model, whose errors may be correlated with the first; the first screener adjudicated. The objective subset gave a result in the same direction.

One model, prompt, option order and seed set. The runs share a preference against option A, so their errors are not independent; shuffling the options per run, a second seed set and a second model are the obvious next tests.

Statistics and literature. Only RQ1 is confirmatory; other analyses are not corrected for multiple comparisons. Fewer items were discordant than the power calculation assumed, so the estimate may be inflated, and the management subgroup cannot detect gains below roughly 9–10 points.

The 3-point threshold is our own choice. Several cited studies were read only as abstracts, and we did not review the literature on benchmark contamination, selective prediction, answer-position bias or GPU non-determinism.

8 Conclusion

On oncology exam questions, taking the majority of ten sampled answers from a small open model was more accurate than a single greedy answer, by a modest and uncertain margin (3.5 points, 95% CI 0.9 to 6.0) that comes mostly from voting and costs 10.0 times the output tokens. The model's answers varied from run to run at its default sampling settings, and their agreement was only a weak signal of correctness, blind to systematic errors. Studies that compare LLMs with tumor boards should report how often each case was run and with which decoding settings. Shuffled option orders, other open models and free-text recommendations on real board cases are the natural next tests.

AI-Use Disclosure

This study was carried out with extensive help from an AI assistant, Claude Code (Anthropic; Claude Opus 5.5), as part of a course on AI tools for research. The assistant proposed candidate gaps and designs, wrote the code, ran the GPU jobs, screened the items and adjudicated disagreements with a second, blind instance of the same model, analysed the results and wrote this manuscript; a further instance that had not written it reviewed the draft, and the assistant revised it after checking each point. The author chose the research gap, design and baseline, approved the frozen protocol and every amendment, and is responsible for the content; these decisions are logged with dates. No clinician reviewed the items or outputs. References were built from DOI, PubMed or arXiv records, but several were read only as abstracts; every number was inserted by a script from the saved results.

Data and Code Availability

Code, configuration files (including the protocol frozen at commit `7890add`), item identifiers, screening codes, parsed answers and all derived results are available at https://github.com/vibhum-phd/lab7-oncology-llm-voting. The MedQA question text is not redistributed, because the licence of the original data is not stated; it can be downloaded from the pinned dataset commit [6]. The raw model outputs restate question text and are therefore not released.

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